

Chapter 9:

Infinite Series

Section 9.4: Comparisons of Series

The Direct Comparison Test

The Direct Comparison Test Suppose that Σa_n and Σb_n are series with positive terms.

(i) If Σb_n is convergent and $a_n \leqslant b_n$ for all n, then Σa_n is also convergent.

(ii) If Σb_n is divergent and $a_n \geqslant b_n$ for all n, then Σa_n is also divergent.

In using the Direct Comparison Test, we must have some known series $\sum b_n$ to compare with. Most of the time we use one of the following types of series

- p-Series $\left[\sum \frac{1}{n^p}\right.$ converges of $p > 1$ and diverges if $p \leq 1$.]
- Geometric Series $[\sum ar^n$ converges of $|r| < 1$ and diverges if $|r| \geq 1$.]

EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} \frac{n}{n^3 + 1}$$

EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} \frac{\cos^2 n}{5^n}$$

EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=3}^{\infty} \frac{1}{\sqrt{n}-1}$$

EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=2}^{\infty} \frac{1}{\ln n}$$

Fact: Let c be any positive number, then

$$\ln n \leq n^c, \qquad n \geq k$$

for some integer k.

EXAMPLE

Determine whether the series is convergent or divergent.

16. $\displaystyle\sum_{n=1}^{\infty} \frac{\cos n + 2}{\sqrt{n}}$

Limit Comparison Test

THEOREM 9.13 Limit Comparison Test

If $a_n > 0$, $b_n > 0$, and

$$\lim_{n\to\infty} \frac{a_n}{b_n} = L$$

where L is *finite and positive*, then

$$\sum_{n=1}^{\infty} a_n \quad \text{and} \quad \sum_{n=1}^{\infty} b_n$$

either both converge or both diverge.



EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} \frac{n}{n^3 + 1}$$

EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} \frac{\sqrt{n+2}}{2n^2+n+1}$$

EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} \frac{1+4^n}{1+3^n}$$

EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$$

Limit Comparison Test

THEOREM 9.13 Limit Comparison Test

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$$\lim_{n\to\infty} \frac{a_n}{b_n} = L$$

where L is *finite and positive*, then

$$\sum_{n=1}^{\infty} a_n \quad \text{and} \quad \sum_{n=1}^{\infty} b_n$$

either both converge or both diverge.



Conclusions: (See Ex. 63 and 64)

1. Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms and $\sum b_n$ is convergent and

$$\lim_{n\to\infty} \frac{a_n}{b_n} = 0$$

then $\sum a_n$ is convergent.

2. Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms and $\sum b_n$ is divergent and

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \infty$$

then $\sum a_n$ is divergent.

EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=2}^{\infty} \frac{1}{\sqrt{n}\,\ln n}$$

EXAMPLE

Determine whether the series is convergent or divergent.

11. $\sum_{n=0}^{\infty} \frac{1}{n!}$

EXAMPLE

67. Proof Suppose that Σa_n is a series with positive terms. Prove that if Σa_n converges, then $\Sigma \sin a_n$ also converges.

Solve exercises of section 9.4.

Start with the suggested problems mentioned in the syllabus.